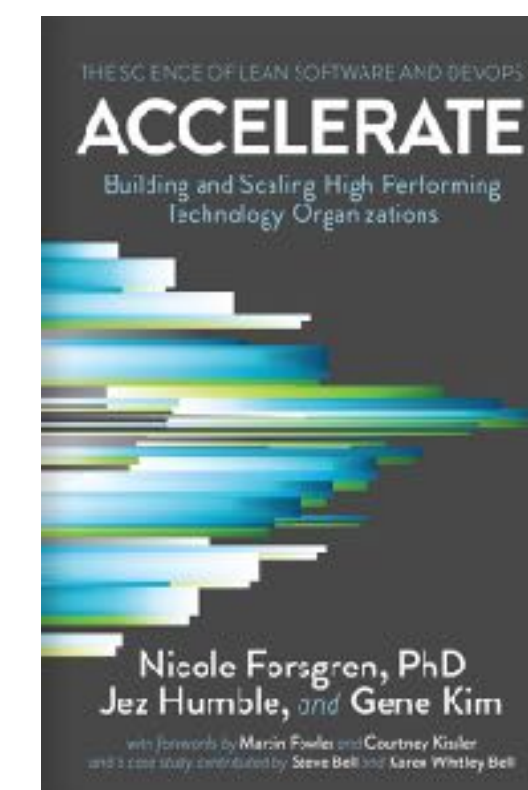
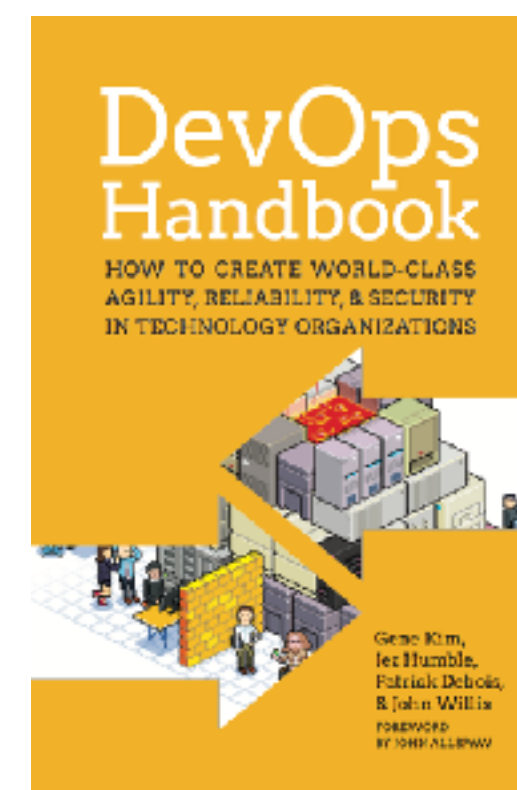
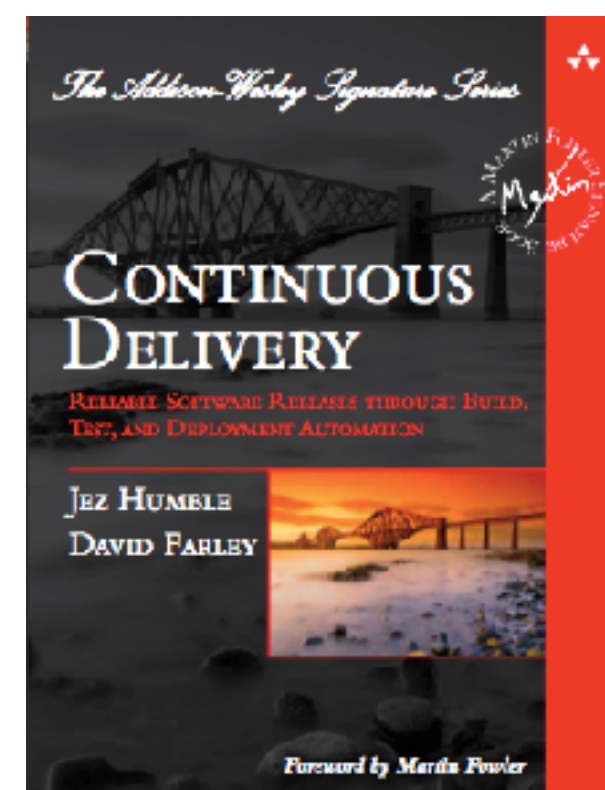




leading a devops transformation

@jezhumble | dev on summit | march 14, 2018



agenda

the problem (why cd matters)

how does devops help?

what devops isn't...

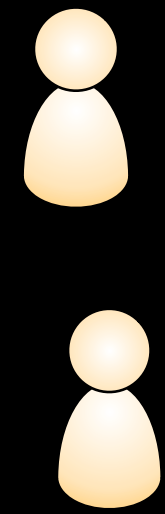
...and what it is

how to get there

“the enterprise”

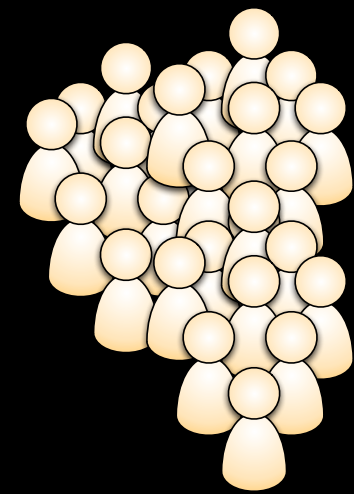
Ping!

Business

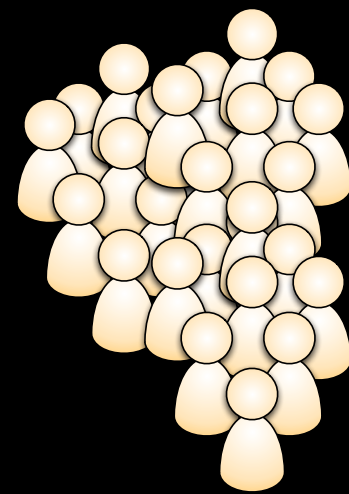


Engineering

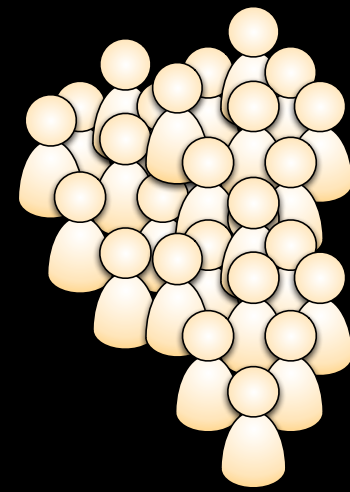
Project C



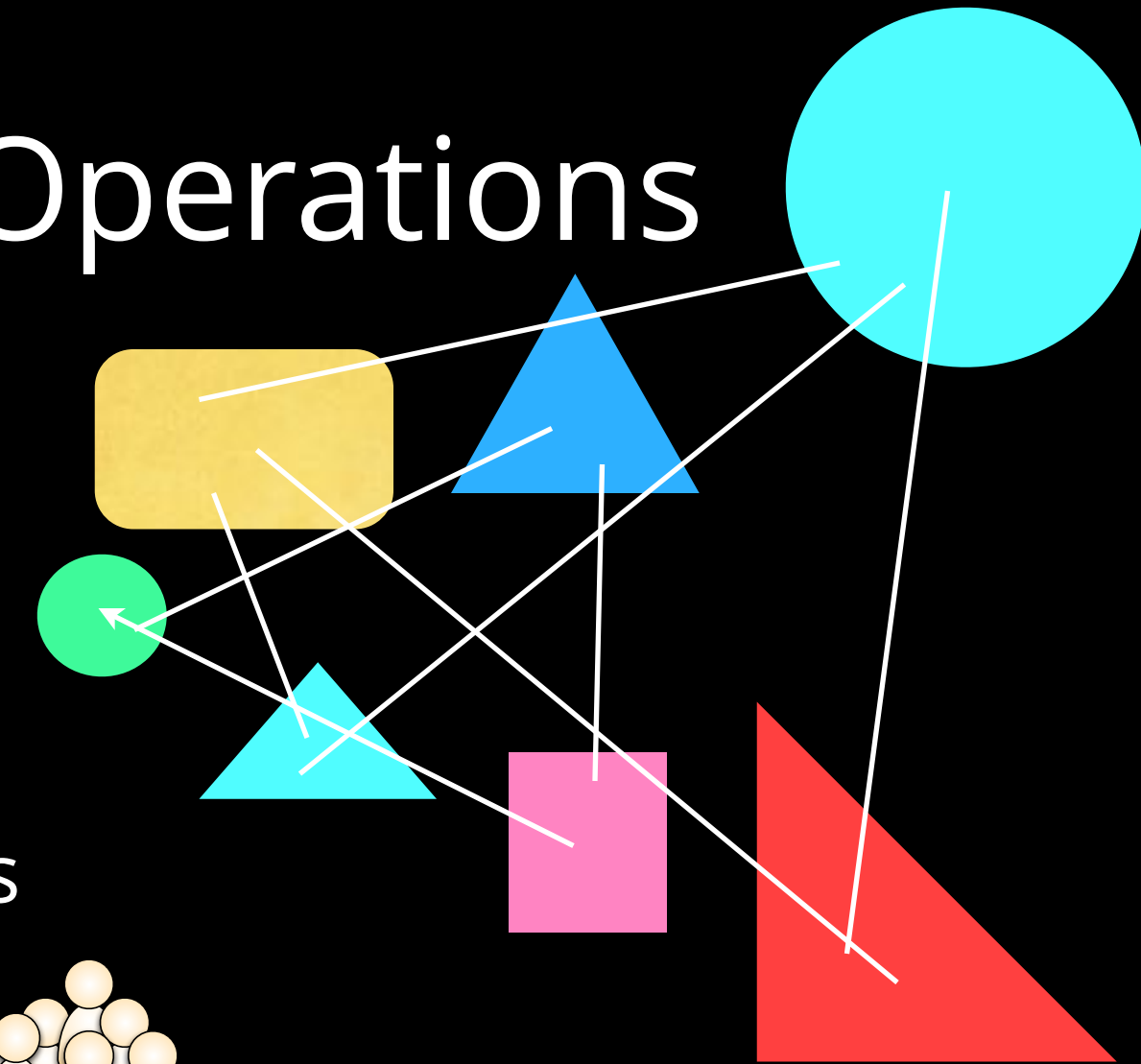
Project A



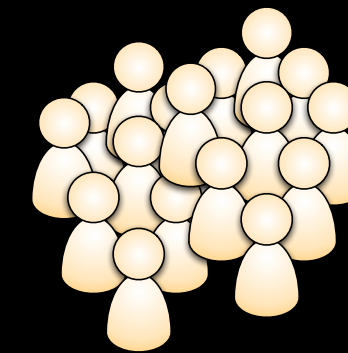
Project B



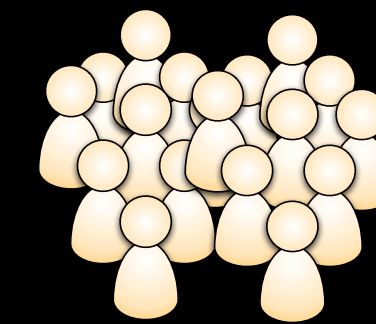
Operations



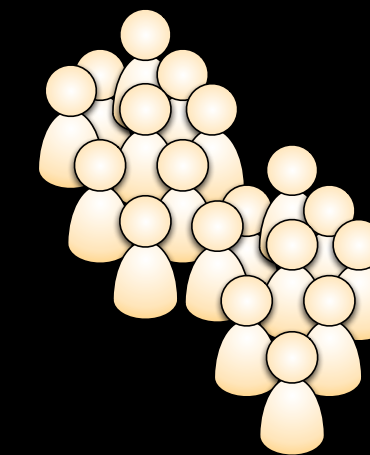
DBAs



Service desk



Infrastructure team



Value stream

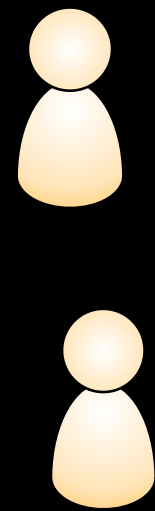


enterprise projects

Ping!

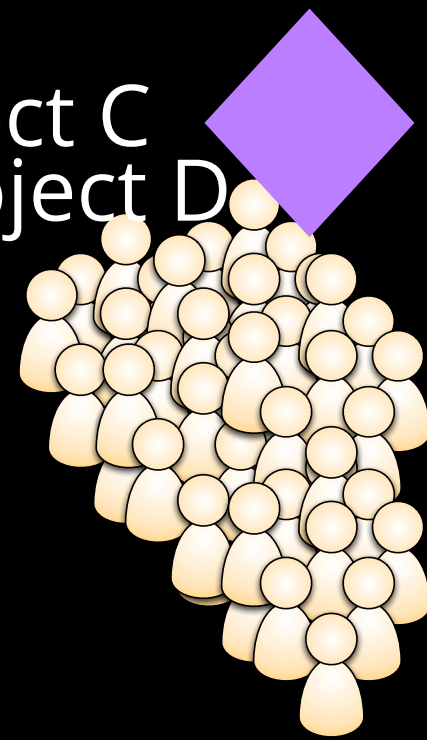
Business

Let's create
a new
product

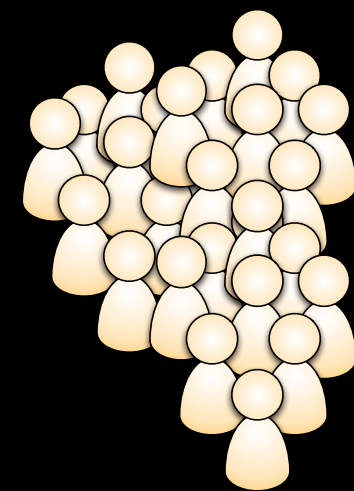
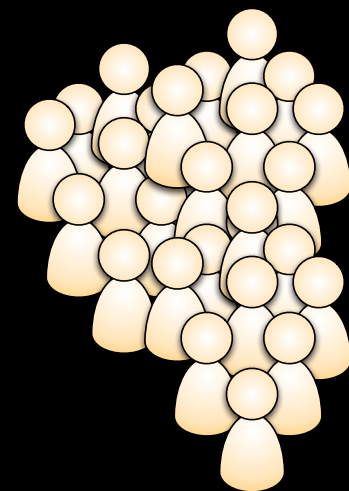


Engineering

Project C
Project D

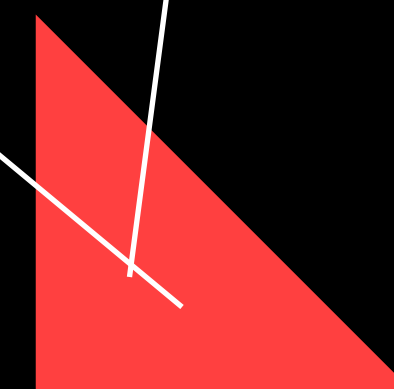
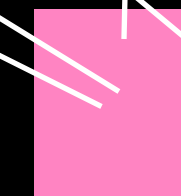
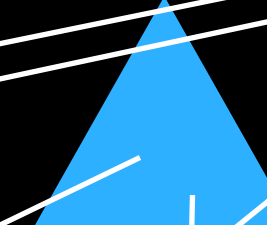
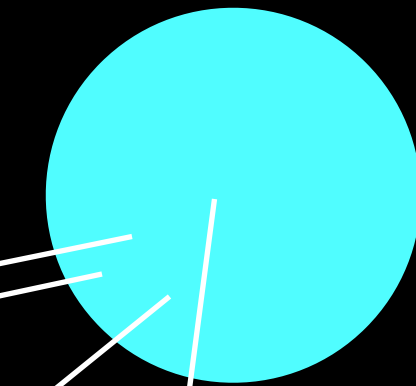


Project A

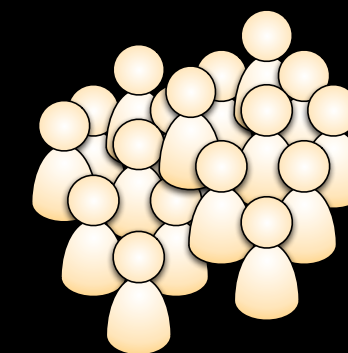


Project B

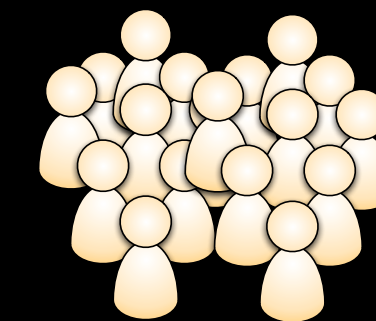
Operations



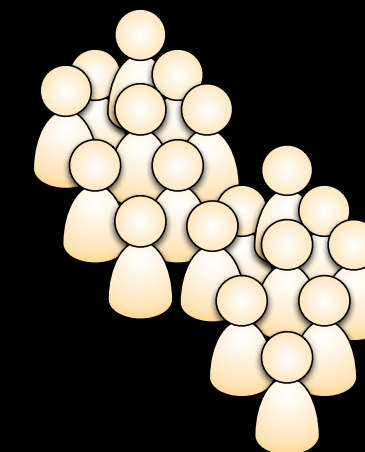
DBAs



Service desk



Infrastructure team

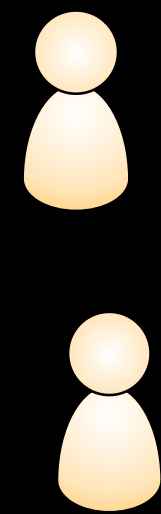


Value stream



Oh no!

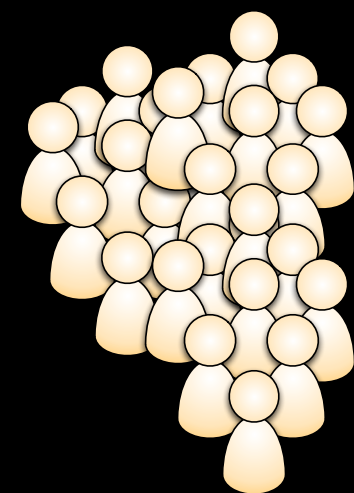
Business



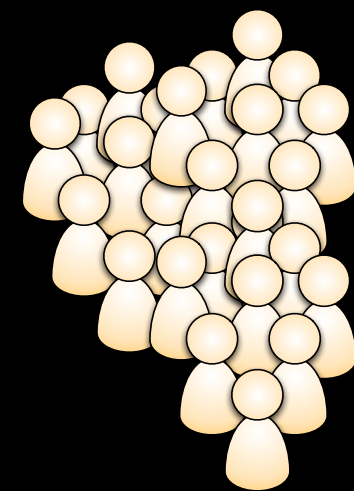
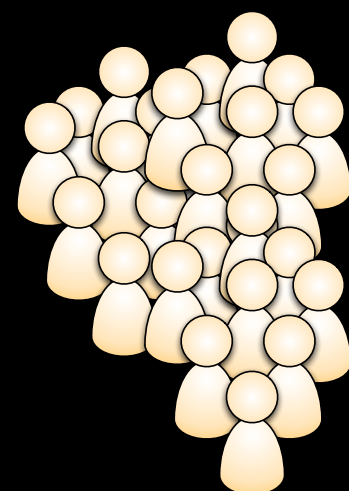
We're going agile!

Engineering

Project D



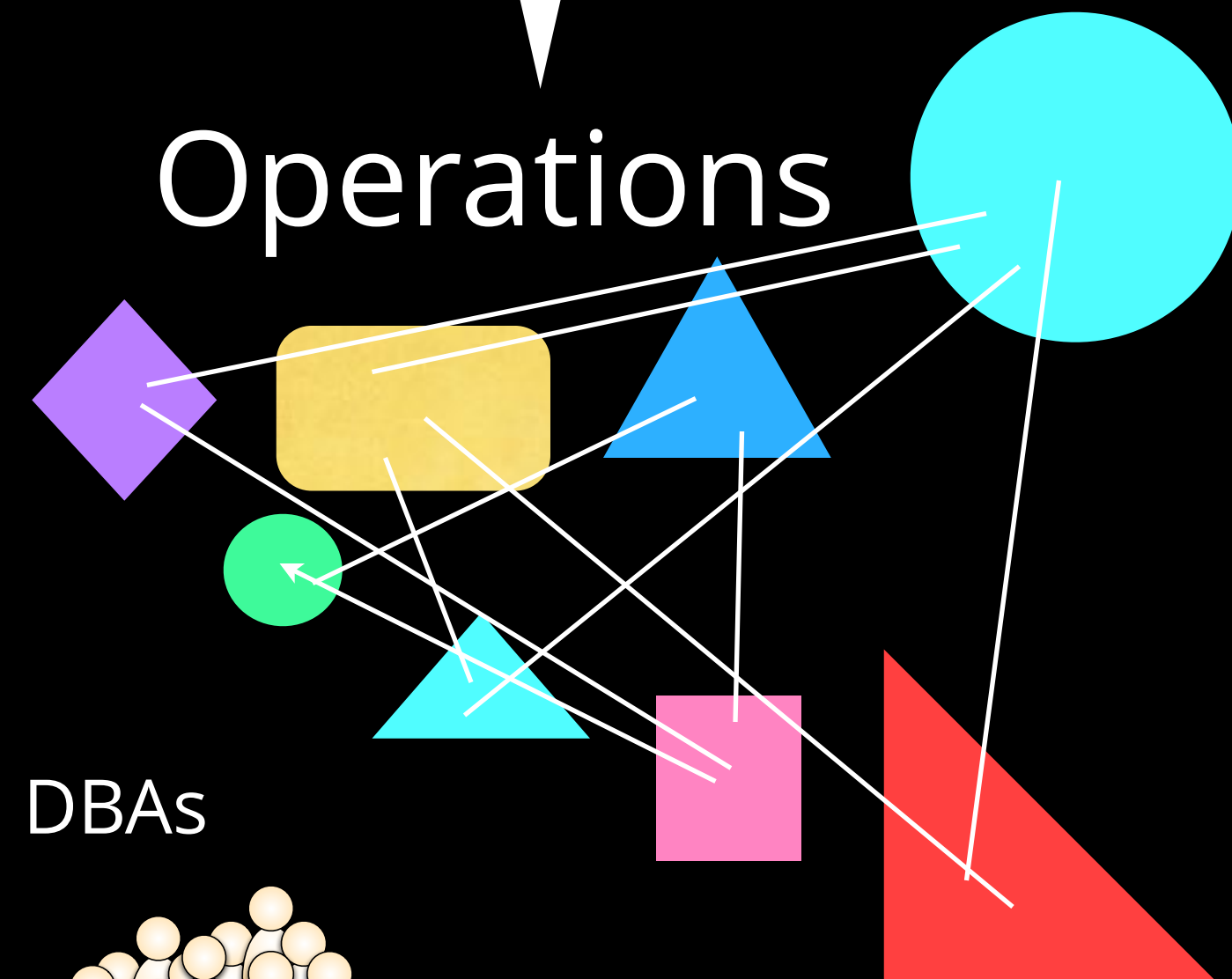
Project A



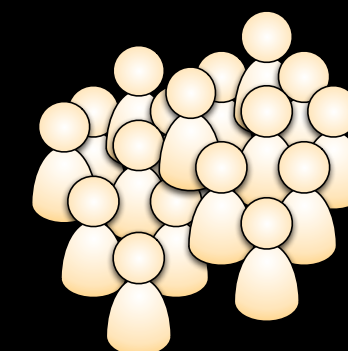
Project B

Oh no!

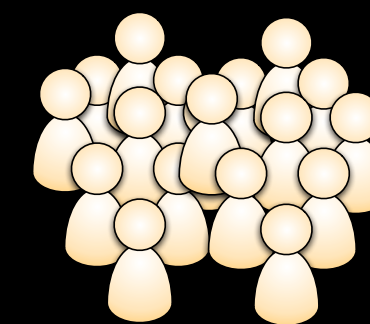
Operations



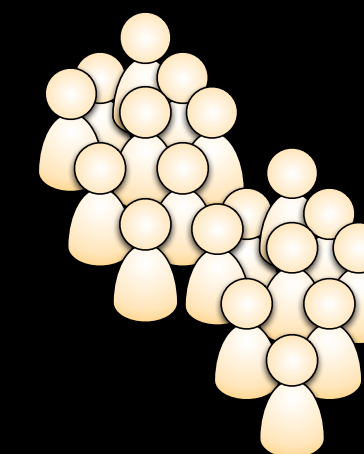
DBAs



Service desk



Infrastructure team



Value stream



Our test-driven code follows SOLID principles

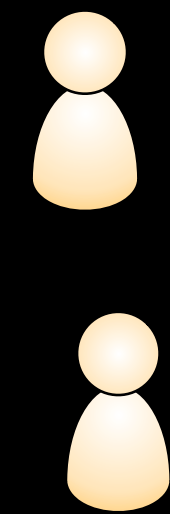
Change management

Shame it doesn't work

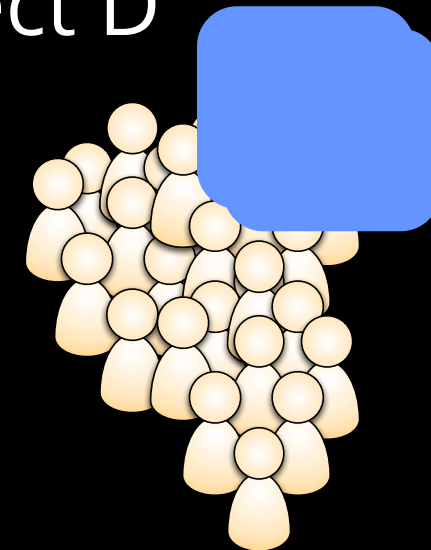
Business

Engineering

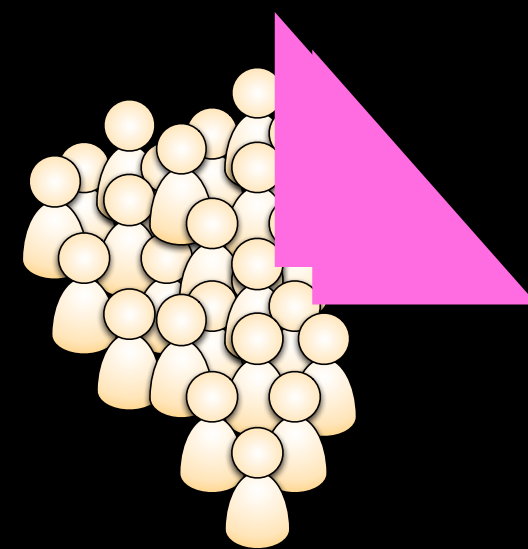
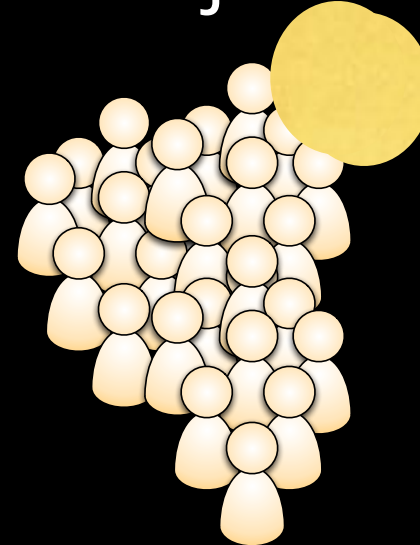
Operations



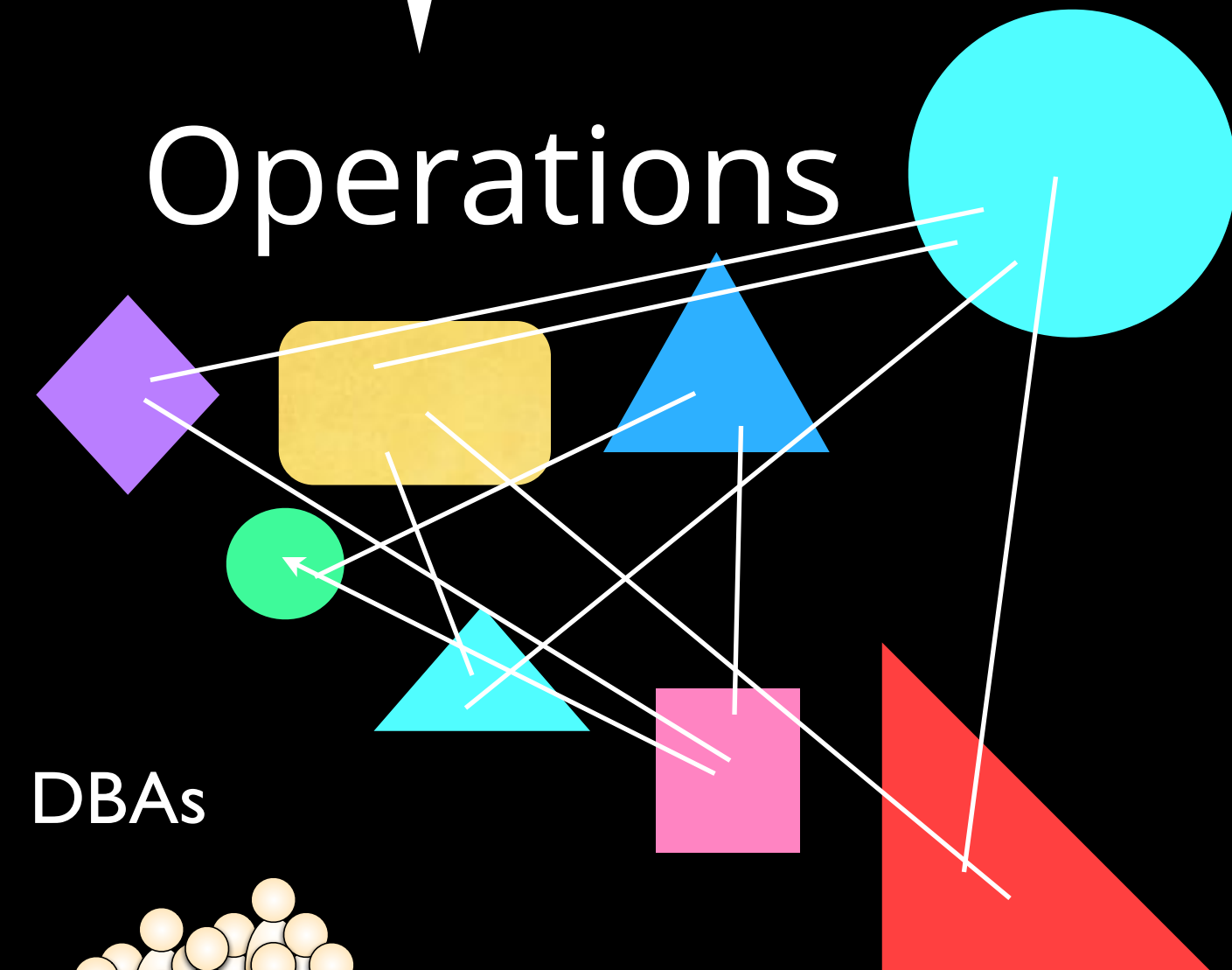
Project D



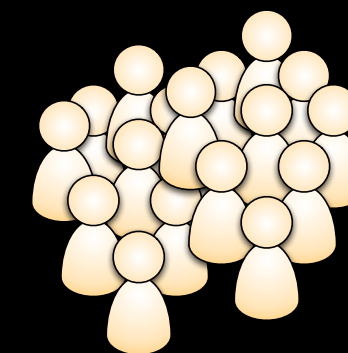
Project A



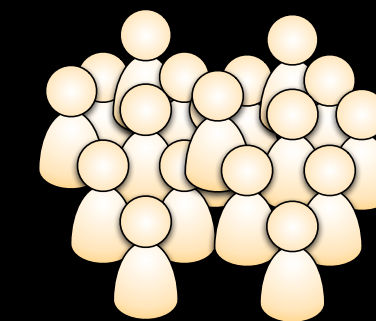
Project B



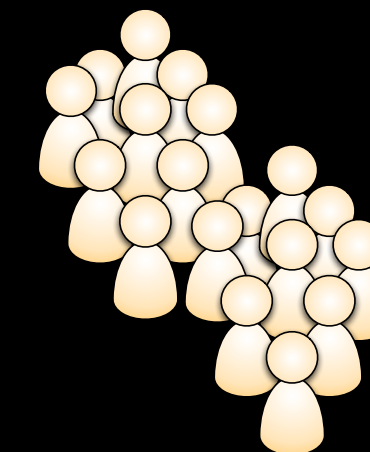
DBAs



Service desk



Infrastructure team



Value stream



how devops helps

devops movement

a cross-functional community of practice dedicated to the study of building, evolving and operating *rapidly changing, secure, resilient systems at scale*

software delivery as a competitive advantage



“Firms with high-performing IT organizations were *twice as likely* to exceed their *profitability, market share* and *productivity* goals.”

<http://bit.ly/2014-devops-report>

software delivery as a competitive advantage

The logo for the 2017 State of DevOps Report. It features a yellow square with the year '2017' in white, followed by the text 'State of DevOps Report' in a bold, black, sans-serif font.

Presented by:



Sponsored by:



<http://bit.ly/2017-devops-report>

high performers were more than twice as likely to achieve or exceed the following objectives:

- Quantity of products or services
- Operating efficiency
- Customer satisfaction
- Quality of products or services provided
- Achieving organizational and mission goals
- Measures that demonstrate to external parties whether or not the organization is achieving intended results

it performance

lead time for changes

release frequency

time to restore service

change fail rate

<http://bit.ly/2014-devops-report>

2016 IT Performance by Cluster

	High IT Performers	Medium IT Performers	Low IT Performers
Deployment frequency <i>For the primary application or service you work on, how often does your organization deploy code?</i>	On demand (multiple deploys per day)	Between once per week and once per month	Between once per month and once every 6 months
Lead time for changes <i>For the primary application or service you work on, what is your lead time for changes (i.e., how long does it take to go from code commit to code successfully running in production)?</i>	Less than one hour	Between one week and one month	Between one month and 6 months
Mean time to recover (MTTR) <i>For the primary application or service you work on, how long does it generally take to restore service when a service incident occurs (e.g., unplanned outage, service impairment)?</i>	Less than one hour	Less than one day	Less than one day*
Change failure rate <i>For the primary application or service you work on, what percentage of the changes either result in degraded service or subsequently require remediation (e.g., lead to service impairment, service outage, require a hotfix, rollback, fix forward, patch)?</i>	0-15%	31-45%	16-30%

* Low performers were lower on average (at a statistically significant level), but had the same median as the medium performers.

<http://bit.ly/2016-devops-report>

Amazon May Deployment Stats

(production hosts & environments only)

11.6 seconds

Mean time between deployments (weekday)

1,079

Max # of deployments in a single hour

10,000

Mean # of hosts simultaneously receiving a deployment

30,000

Max # of hosts simultaneously receiving a deployment

it's about *culture* and *architecture*

not

process, tools, organizational structure

myths: devops *isn't*...

...firing our sysadmins / testers / ... and hiring “devops experts”

...doing a re-org

...rewriting all our systems and sticking them in the cloud

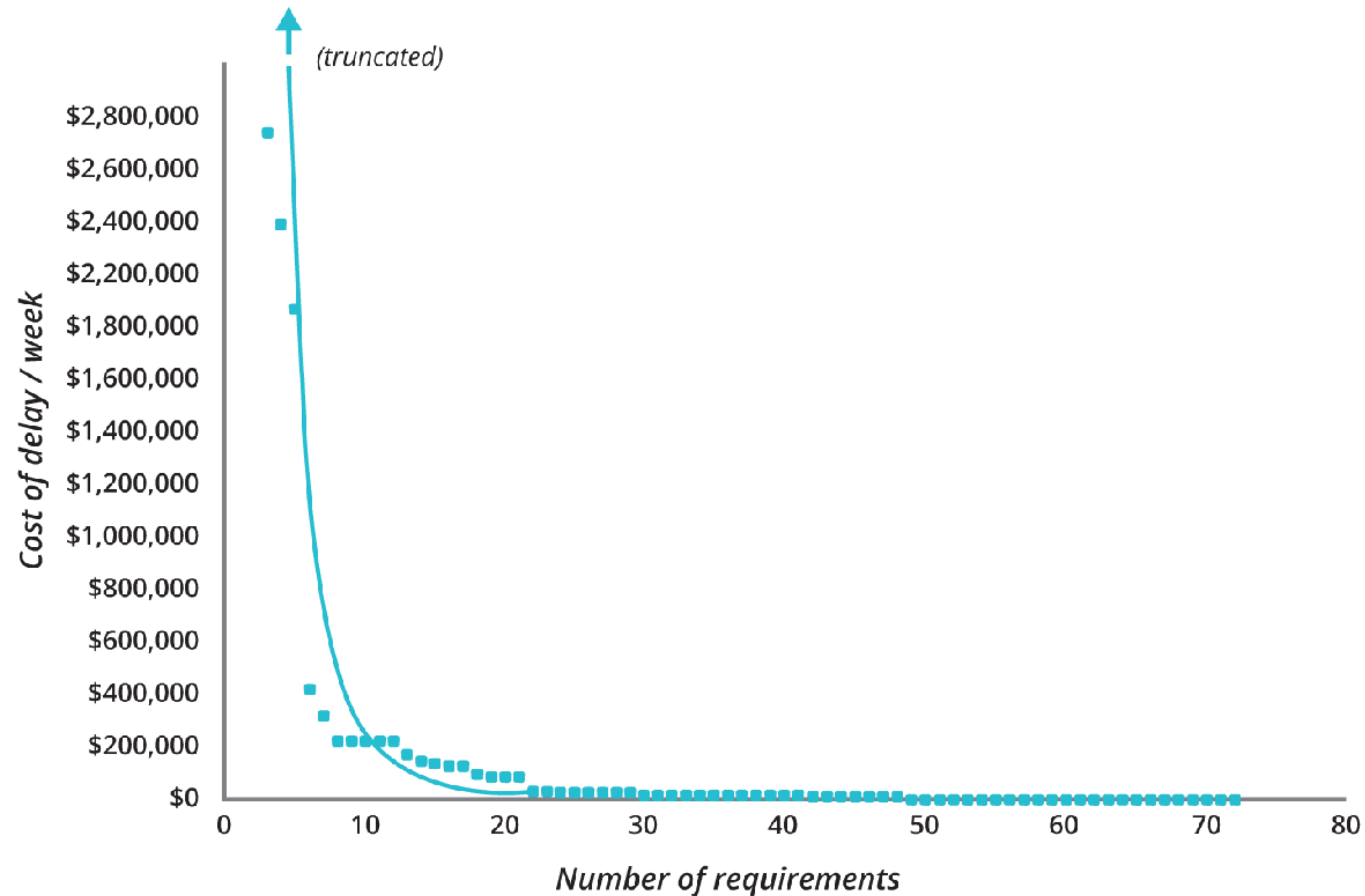
...buying a bunch of devops tools

...giving developers (or anyone else for that matter) access to prod

devops *is...*

...learning to work in small batches (product dev & org change)

projects are evil



architectural outcomes: can my team...

make large-scale changes to the design of its system without the permission of somebody outside the team or depending on other teams?

complete its work without needing fine-grained communication and coordination with people outside the team?

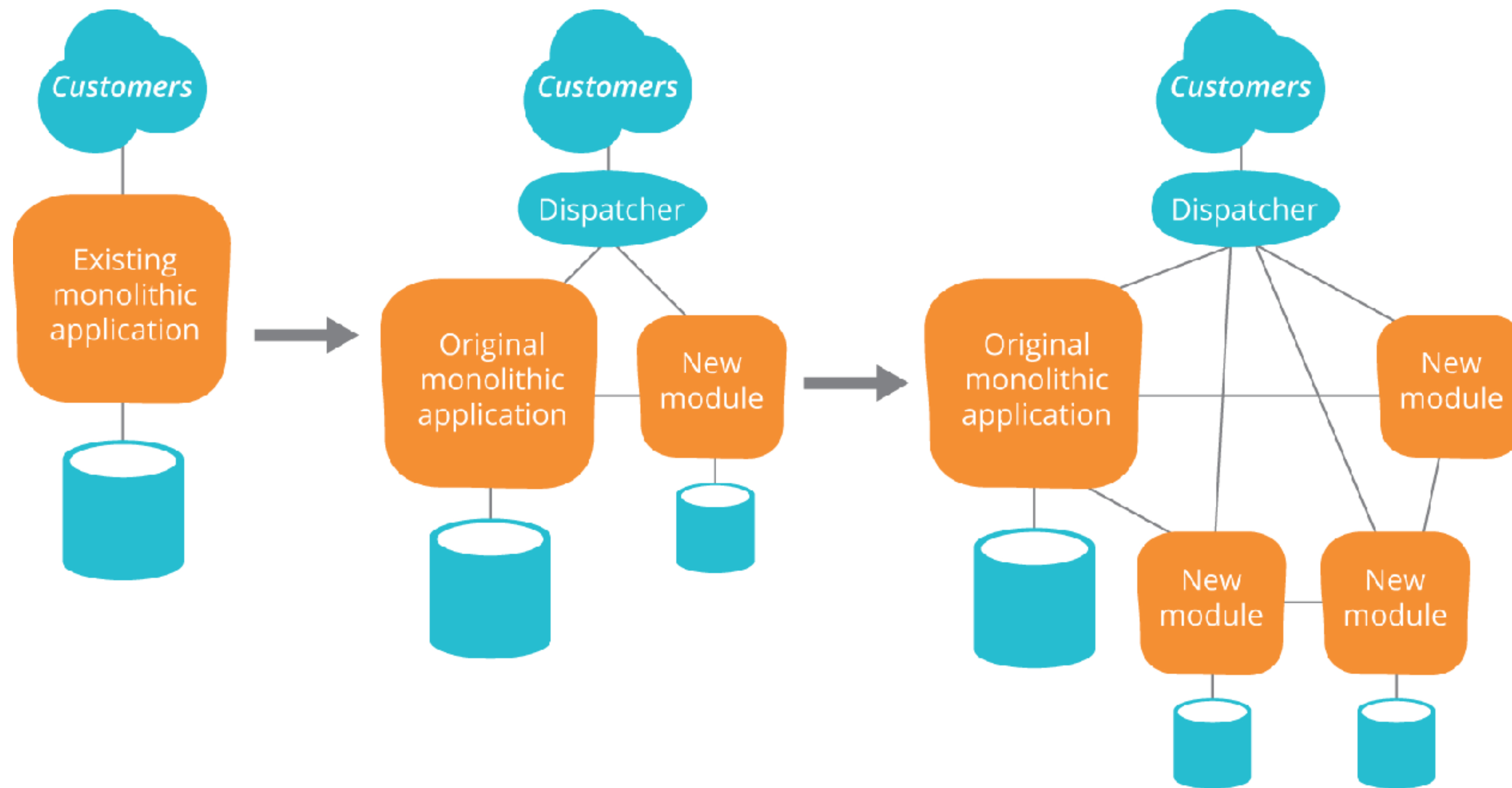
deploy and release its product or service on demand, independently of other services the product or service depends upon?

do most of its testing on demand, without requiring an integrated test environment?

perform deployments during normal business hours with negligible downtime?



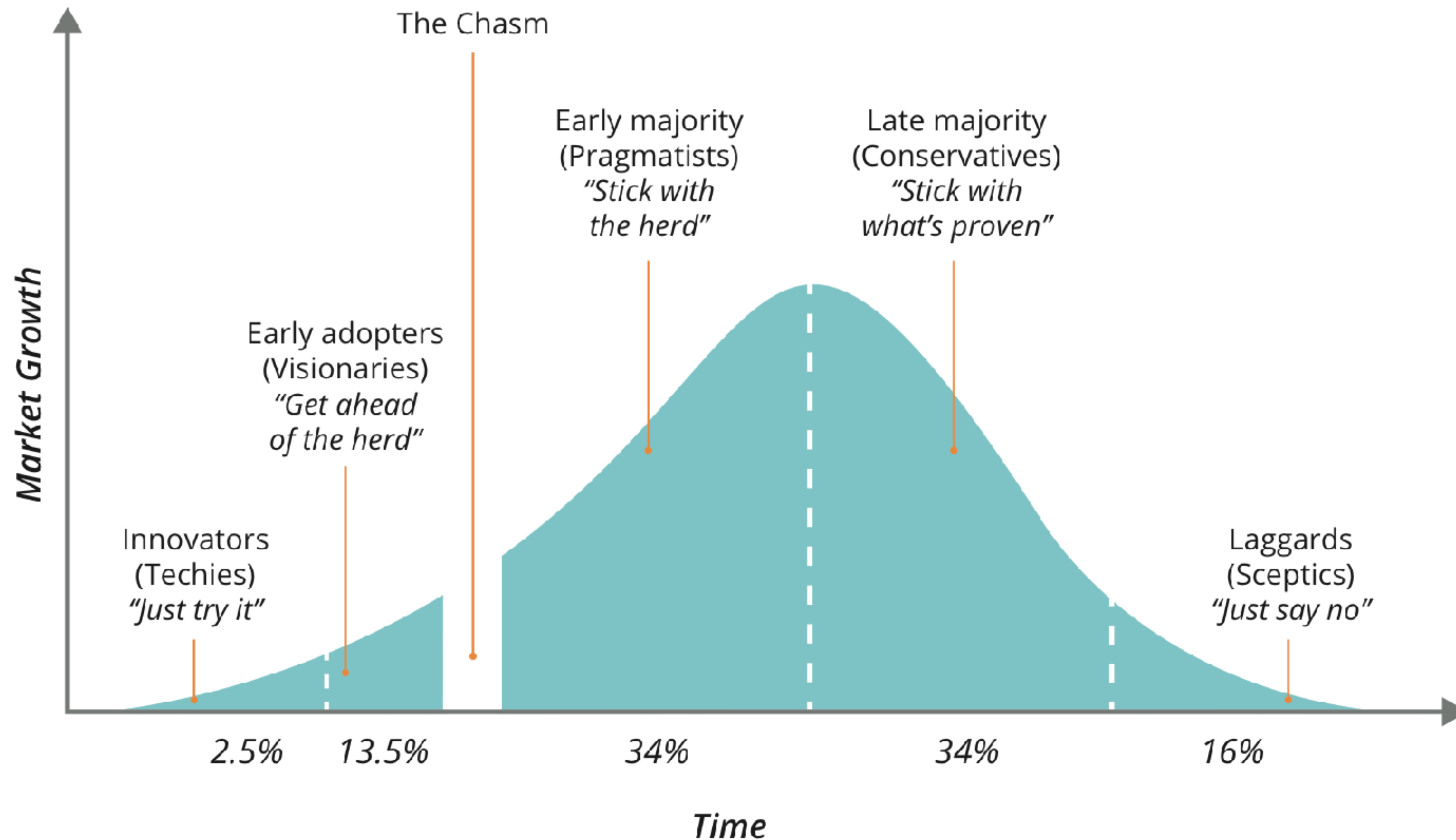
strangler application



rules of strangler

- start by delivering new functionality—at least at first
- don't rewrite existing functionality except to simplify
- deliver something fast
- design for testability and deployability
- architect the new software to run on a paas

org change



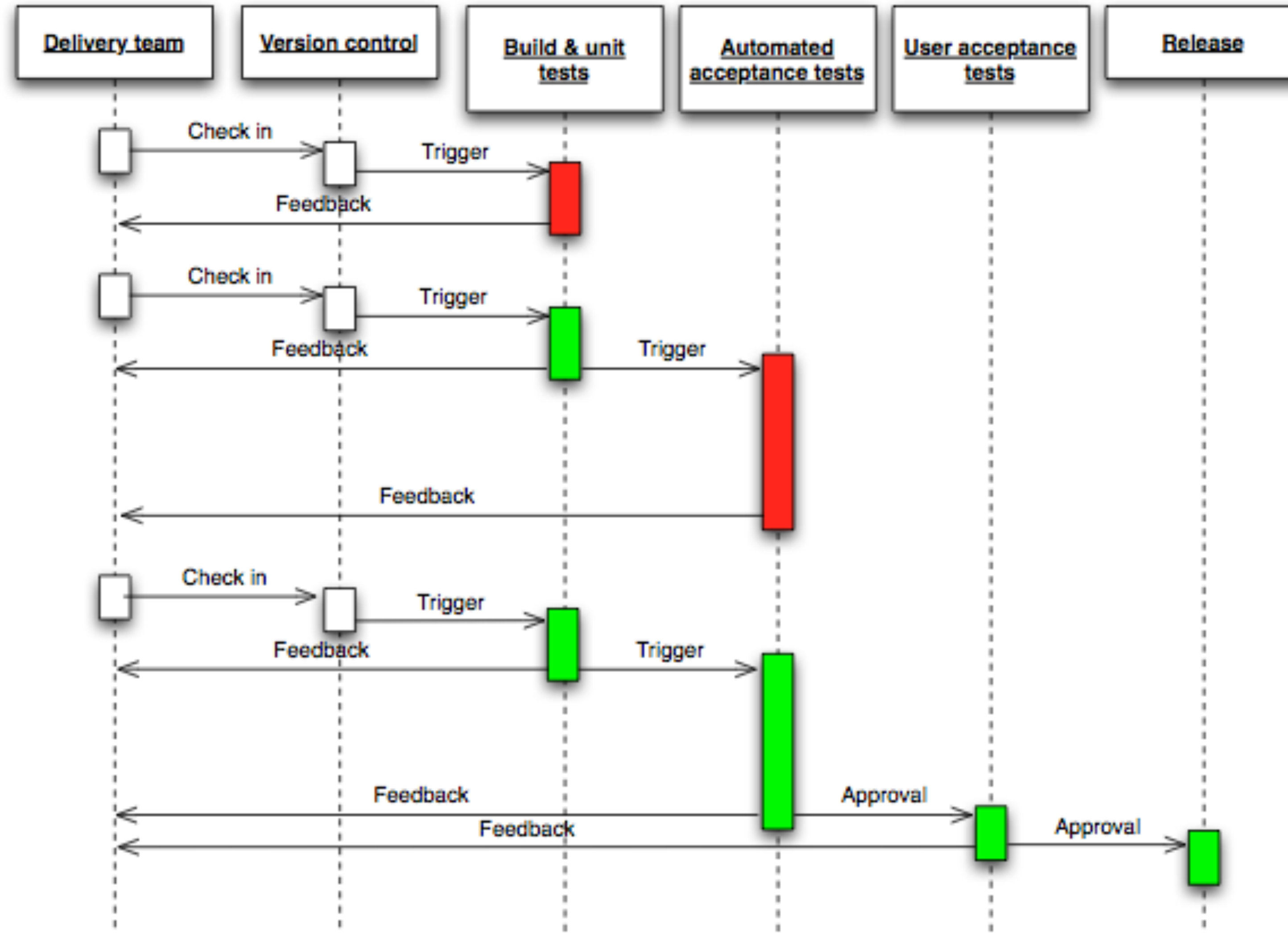
Geoffrey Moore, *Crossing the Chasm*

devops *is...*

...learning to work in small batches (product dev & org change)

...creating fast feedback loops

deployment pipeline



devops *is...*

...learning to work in small batches (product dev & org change)

...creating fast feedback loops

...enabling cross-functional collaboration through value streams

devops *is...*

...learning to work in small batches (product dev & org change)

...creating fast feedback loops

...enabling cross-functional collaboration across value streams

...developing a culture of experimentation

high trust culture

how organizations process information

Pathological (<i>power oriented</i>)	Bureaucratic (<i>rule oriented</i>)	Generative (<i>performance oriented</i>)
Low cooperation	Modest cooperation	High cooperation
Messengers shot	Messengers neglected	Messengers trained
Responsibilities shirked	Narrow responsibilities	Risks are shared
Bridging discouraged	Bridging tolerated	Bridging encouraged
Failure leads to scapegoating	Failure leads to justice	Failure leads to enquiry
Novelty crushed	Novelty leads to problems	Novelty implemented

effective teams



<https://rework.withgoogle.com/blog/five-keys-to-a-successful-google-team/>

disaster recovery testing

“For DiRT-style events to be successful, an organization first needs to accept system and process failures as a means of learning... We design tests that require engineers from several groups who might not normally work together to interact with each other. That way, should a real large-scale disaster ever strike, these people will already have strong working relationships”

—Kripa Krishnan, Director, Cloud Operations, Google

A photograph of two men on a stage. The man on the left, wearing a plaid shirt, is gesturing with his right hand while holding a piece of paper. The man on the right, wearing a textured sweater, is looking towards him. In the background, a large screen displays the word "Etsy" in a stylized font. To the left of the screen, the text "Three Arrows and Sweater" is visible. A red "EXIT" sign is mounted on the wall to the far left. A dark blue text box is overlaid on the image, containing white text.

The immediate response
from *everyone* around was to ask, "What help
do you need?"

NETFLIX



devops *is...*

...learning to work in small batches (product dev & org change)

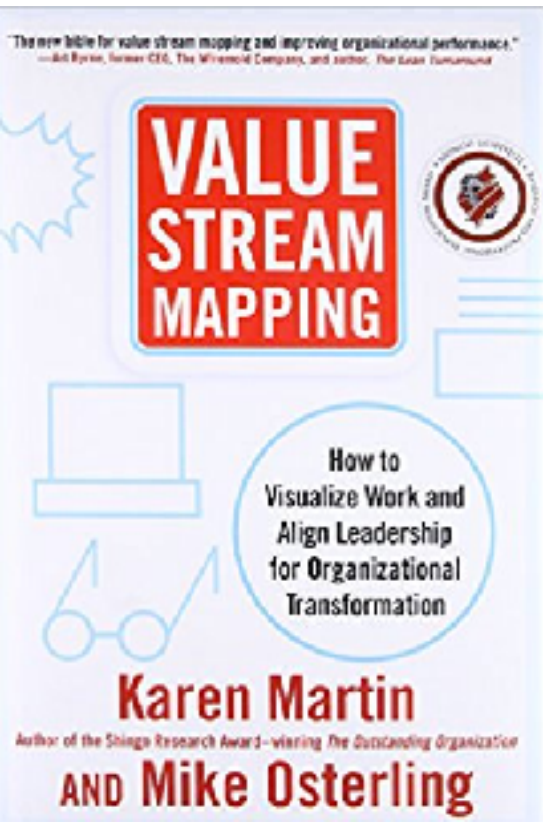
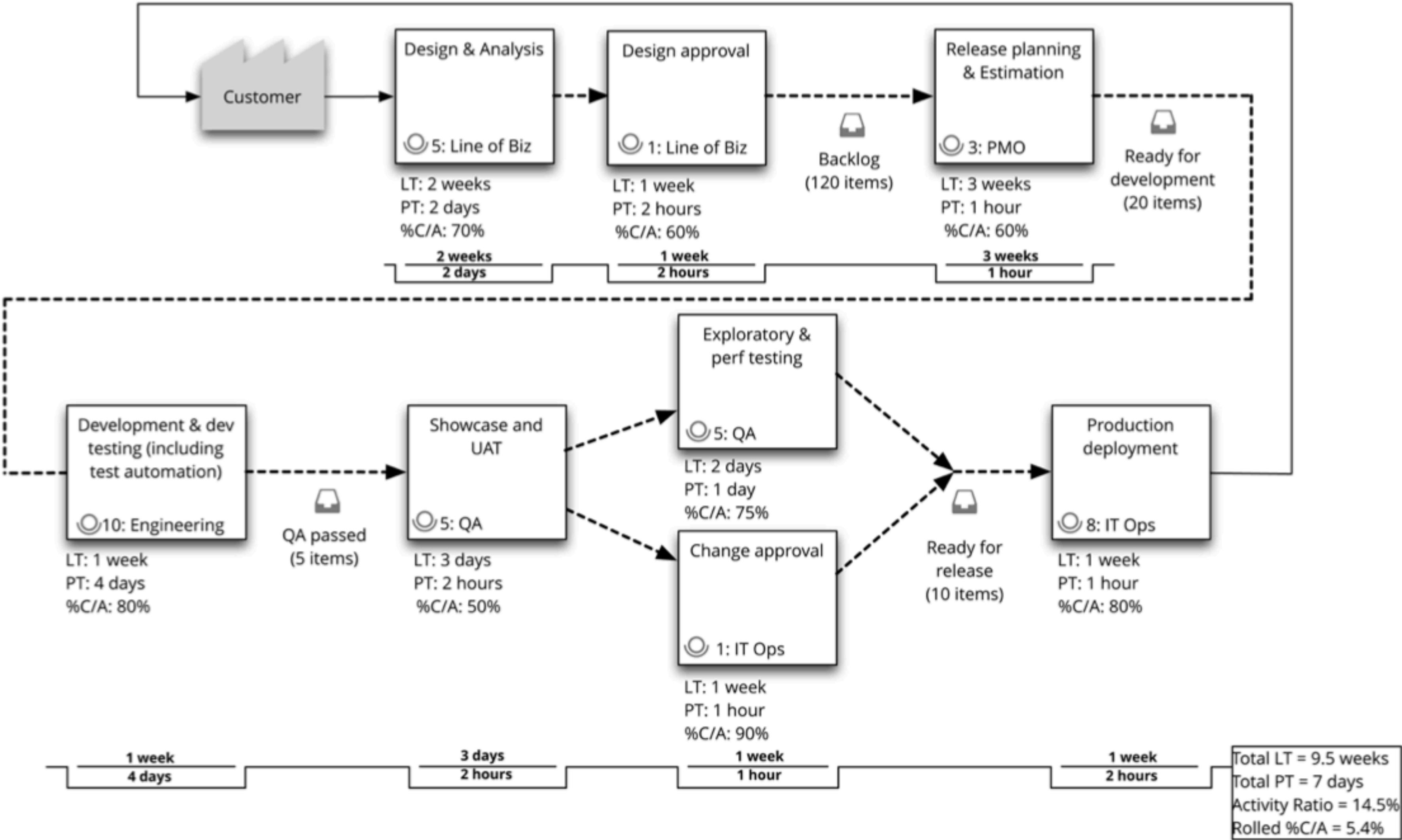
...creating fast feedback loops

...enabling cross-functional collaboration across value streams

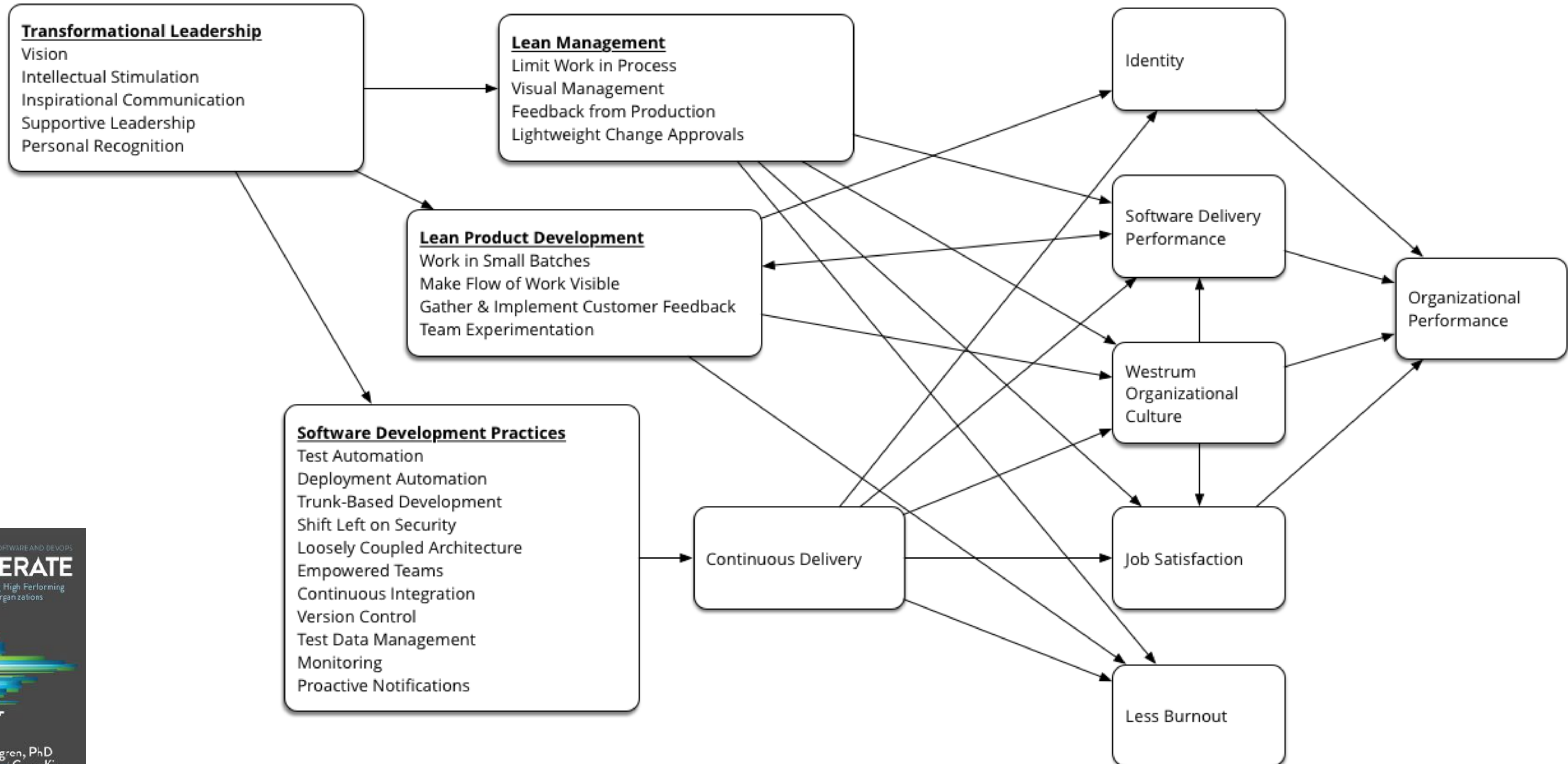
...making it safe to fail

...continually working to remove waste and make things better

value stream mapping

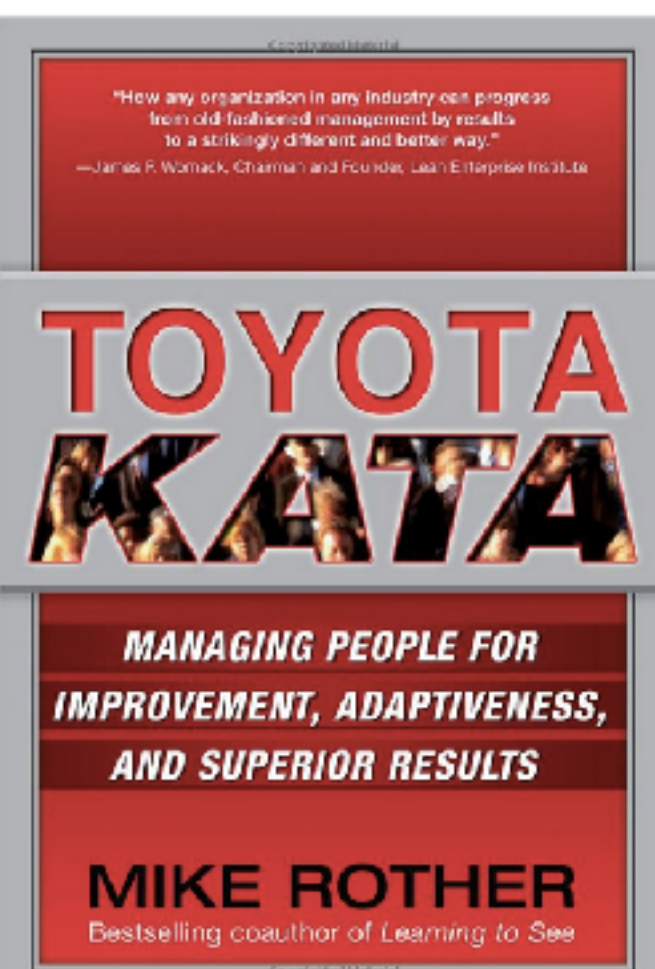
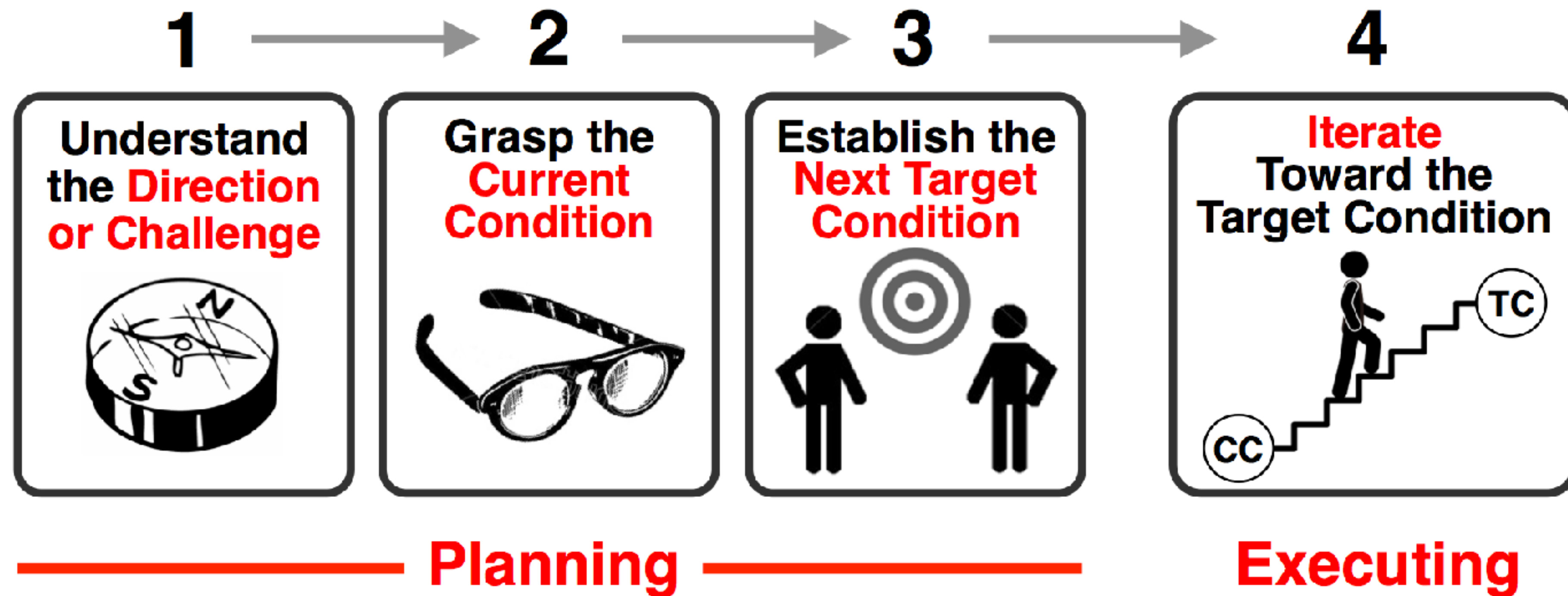


improving performance



THE FOUR STEPS OF THE IMPROVEMENT KATA MODEL

A systematic, scientific pattern of working



Mike Rother | <http://www-personal.umich.edu/~mrother/>

the journey

agree and communicate measurable business goals

give teams resources to experiment and reward learning

encourage collaboration

achieve quick wins and share lessons learned

keep going

thank you!

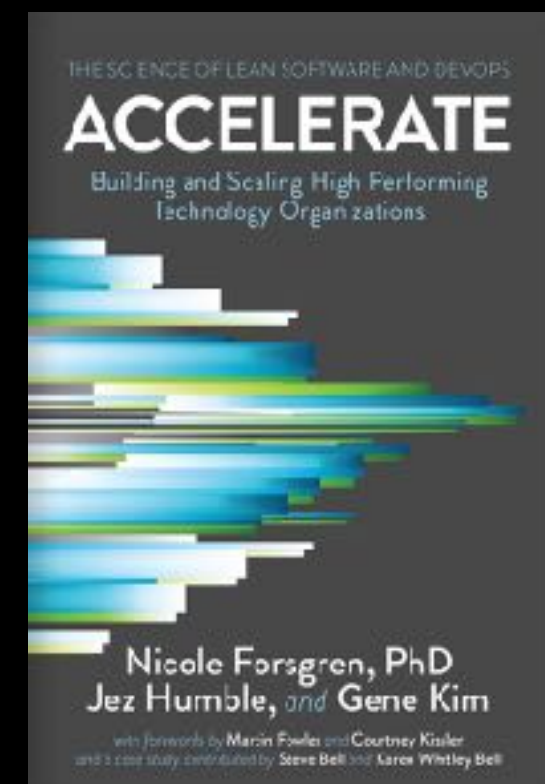
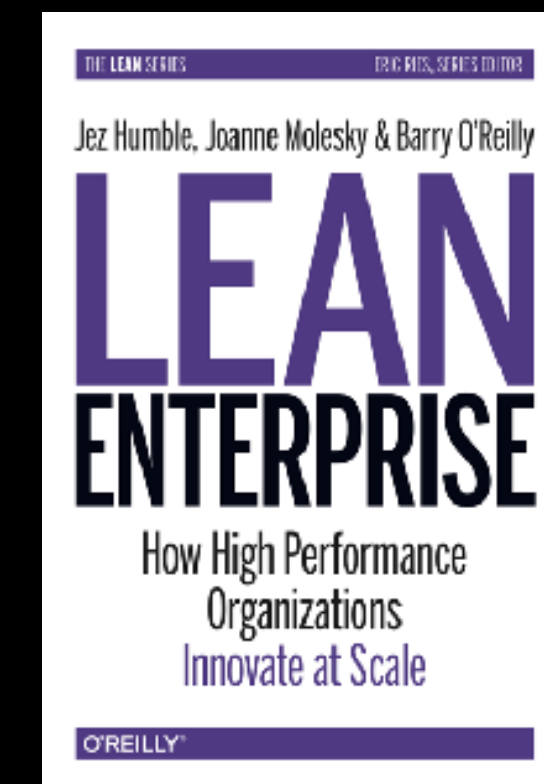
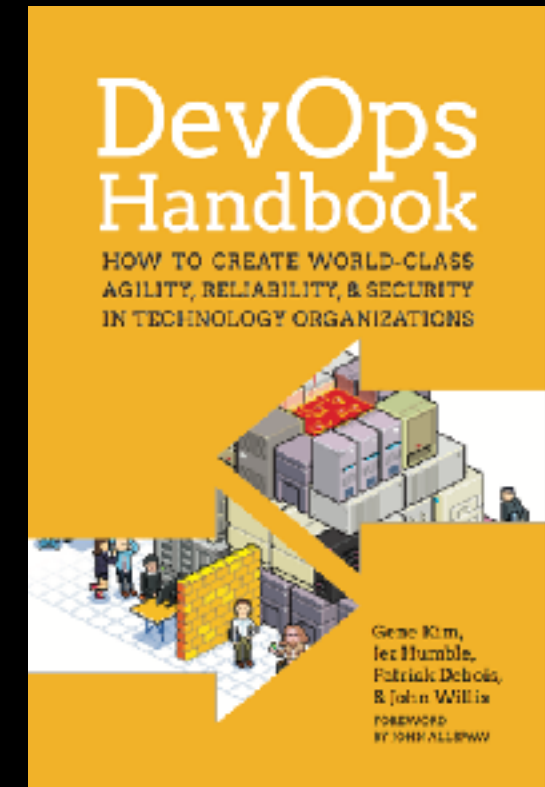
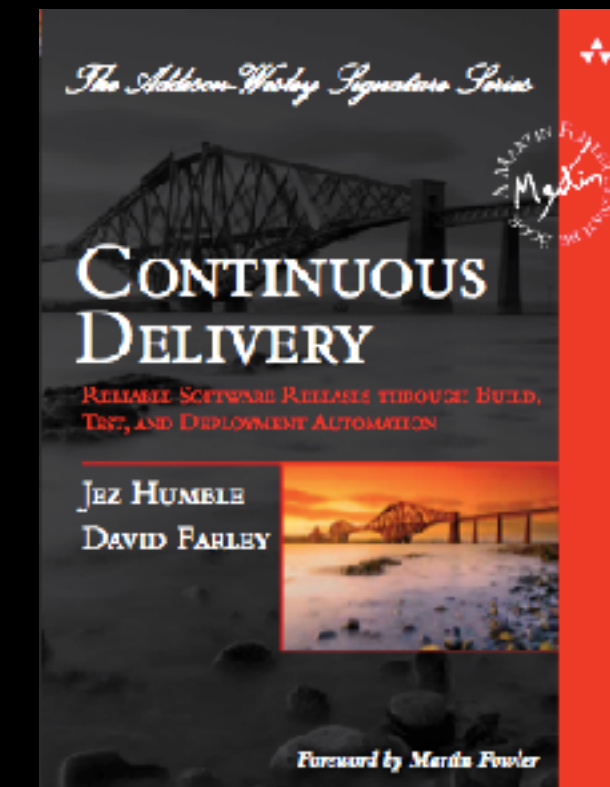
To receive the following:

- A copy of this presentation
- A 100 page excerpt from *Lean Enterprise*
- Excerpts from the *DevOps Handbook* and *Accelerate*
- A 20m preview of my Continuous Delivery video workshop
- Discount code for CD video + interviews with Eric Ries & more

Just pick up your phone and send an email

To: jezhumble@sendyourslides.com

Subject: devops



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